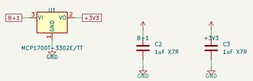
A — CP304 power interface (B+1 in, B+3 out)

J1 reuses the CP304 socket: pin 1 = B+1 (6.0 V from Power Board via S901), 2 = GND, 3 = B+3 to Q703/Q704.
No series diode on B+1 (guide 5). Connector family TBD.



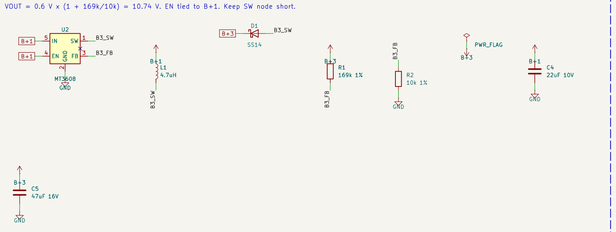
B — Local +3V3 (MCP1700, 250 mA max)

Datasheet: 1 uF ceramic in and out. Not powered from SWD or USB VBUS.



C — B+3 motor rail (MT3608 boost, ~10.8 V)

VOUT = 0.6 V × (1 + 169k/10k) = 10.74 V. EN tied to B+1. Keep SW node short.

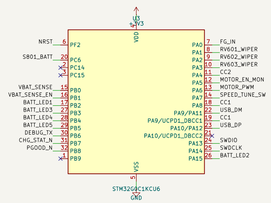


I — Power Board telemetry

VBAT_SENSE arrives ADC-safe (divided on Power Board). CHG/PGOOD are BQ24074 open-drain, active low.

D — STM32G0C1KCU6 + SWD

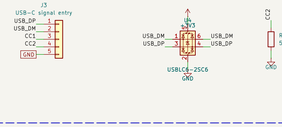
Pin allocation per guide 8 (rev 2026-11-09). BOOT0/PA14 10k pull-down, NRST 100 nF.
PA8 = UCPD1_CC1, PA9 = UCPD1_DBCC1 tied to CC1 (RM0444 7.3.16). CC2 -> PA4 (ADC_IN4).





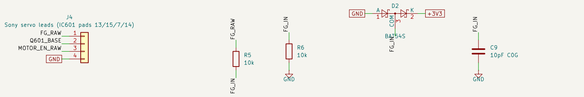
E — USB-C signalling (5 V sink, no PD)

J3 is a placeholder until the USB-C receptacle location is decided (handoff D1).
CC1 Rd is the STM32's own (dead-battery Rd unpowered, UCPD sink Rd once firmware runs); no external resistor. CC2 has no UCPD pin on this package: external 5.1k Rd, read by ADC on PA4.
Firmware: set UCPD1 sink on CC1 before UCPD1_STROBE; never drive PA9. USBLC6 VBUS pin -> +3V3.



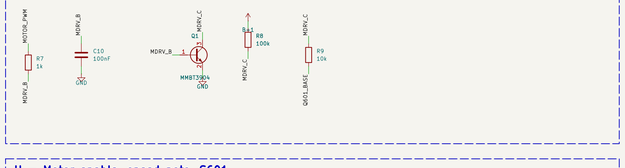
F — FG input (FG901 open collector)

PROVISIONAL: 10k/10k keeps a 6 V FG high at 3.0 V without clamp conduction.
Final values from the measured swing at CX20084 pad 13.



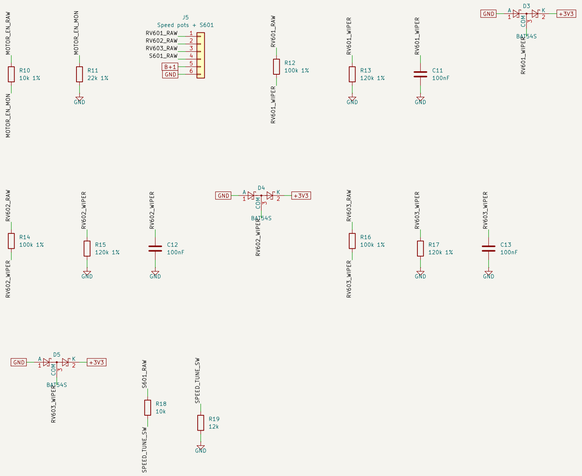
G — Motor drive to Q601 base

PA6 PWM -> 1k/100nF filter -> MMBT3904 -> 10k -> Q601 base; 100k pull-up to B+1 = safe off.
PROVISIONAL until Q601 is identified and characterised (R and C values, sign).



H — Motor enable, speed pots, S601

MOTOR_EN: 4.4 V x 22k/32k = 3.03 V at PA5. Pots: 100k/120k caps a 6 V wiper at 3.27 V:
100 nF at each ADC pin. All PROVISIONAL until wiper ranges are measured.

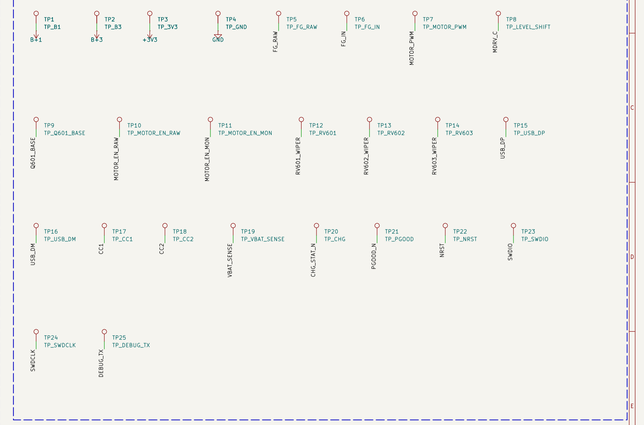


J — LED battery monitor (mandatory)

Outputs must be released (high-Z) outside BATT mode. 1k series and SB01 divider PROVISIONAL
until LED polarity. CX10043 behaviour and SB01 level are measured; buffers may be needed.



Test points (guide 17)



Connector families/pitch not yet chosen: generic connectors, empty footprints
PROVISIONAL values await bench measurement on WM-D6C SN72795 (PCB C11-494-12)
Generated by tools/gen_servo_board.py; authority: docs/hardware/kicad-servo-board-schematic-guide.md
wmd6c-servo (CERN-OHL-P v2)

Sheet: /	
File: DSR-1.kicad_sch	
Title: DSR-1 Servo Control Board	
Size: D	Date: 2026-11-09
KiCad E.D.A. 10.0.8	Id: 1/1

Rev: A-draft1